Test Series Code

15

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 22/04/2026

ANSWER KEY

1.	(2)	46.	(2)	91.	(3)	136.	(3)
2.	(1)	47.	(2)	92.	(4)	137.	(3)
3.	(2)	48.	(3)	93.	(2)	138.	(4)
4.	(2)	49.	(3)	94.	(1)	139.	(1)
5.	(1)	50.	(1)	95.	(4)	140.	(1)
6.	(2)	51.	(1)	96.	(3)	141.	(2)
7.	(2)	52.	(3)	97.	(3)	142.	(3)
8.	(3)	53.	(1)	98.	(1)	143.	(2)
9.	(3)	54.	(2)	99.	(3)	144.	(3)
10.	(1)	55.	(1)	100.	(4)	145.	(4)
11.	(4)	56.	(2)	101.	(3)	146.	(3)
12.	(1)	57.	(3)	102.	(2)	147.	(3)
13.	(4)	58.	(3)	103.	(3)	148.	(1)
14.	(2)	59.	(2)	104.	(3)	149.	(2)
15.	(1)	60.	(3)	105.	(3)	150.	(3)
16.	(2)	61.	(2)	106.	(3)	151.	(3)
17.	(4)	62.	(1)	107.	(3)	152.	(3)
18.	(4)	63.	(4)	108.	(2)	153.	(2)
19.	(2)	64.	(3)	109.	(4)	154.	(2)
20.	(4)	65.	(3)	110.	(4)	155.	(2)
21.	(2)	66.	(2)	111.	(4)	156.	(1)
22.	(1)	67.	(2)	112.	(3)	157.	(2)
23.	(4)	68.	(1)	113.	(4)	158.	(3)
24.	(4)	69.	(3)	114.	(3)	159.	(4)
25.	(3)	70.	(2)	115.	(3)	160.	(1)
26.	(3)	71.	(2)	116.	(3)	161.	(3)
27.	(3)	72.	(1)	117.	(2)	162.	(3)
28.	(1)	73.	(1)	118.	(3)	163.	(4)
29.	(4)	74.	(1)	119.	(2)	164.	(3)
30.	(3)	75.	(2)	120.	(4)	165.	(3)
31.	(3)	76.	(3)	121.	(2)	166.	(2)
32.	(2)	77.	(2)	122.	(4)	167.	(3)
33.	(4)	78.	(1)	123.	(2)	168.	(1)
34.	(1)	79.	(3)	124.	(2)	169.	(1)
35.	(1)	80.	(2)	125.	(3)	170.	(2)
36.	(2)	81.	(3)	126.	(2)	171.	(2)
37.	(2)	82.	(4)	127.	(1)	172.	(3)
38.	(4)	83.	(2)	128.	(4)	173.	(1)
39.	(2)	84.	(4)	129.	(4)	174.	(3)
40.	(3)	85.	(1)	130.	(1)	175.	(1)
41.	(1)	86.	(1)	131.	(3)	176.	(1)
42.	(1)	87.	(1)	132.	(4)	177.	(2)
43.	(2)	88.	(2)	133.	(2)	178.	(4)
44.	(3)	89.	(1)	134.	(4)	179.	(1)
45.	(2)	90.	(1)	135.	(1)	180.	(1)

Hints & Solutions

1. (2)

For the distance r to always increase, the velocity vector ($\vec{v}$) must always have a component pointing away from the origin ($\vec{r}$).

$\vec{r}\cdot\vec{v}\geq 0$ for all time t.

Using position $\vec{r}=(ut\cos\theta, ut\sin\theta-\frac{1}{2}gt^2)$ and velocity $\vec{v}=(u\cos\theta, u\sin\theta-gt)$:

$$\vec{r}\cdot\vec{v}=u^2t-\frac{3}{2}ugt^2\sin\theta+\frac{1}{2}g^2t^3\geq 0$$

Dividing by t gives a quadratic in t:

$$\frac{1}{2}g^2t^2-\frac{3}{2}ug\sin\theta\cdot t+u^2\geq 0$$

$$g^2t^2-3ug\sin\theta t+2u^2\geq 0$$

$$g^2\left(t^2-\frac{3u\sin\theta}{g}t\right)+2u^2\geq 0$$

Complete the square inside by adding and subtracting $\left(\frac{3u\sin\theta}{2g}\right)^2$:

$$g^2\left[\left(t-\frac{3u\sin\theta}{2g}\right)^2-\frac{9u^2\sin^2\theta}{4g^2}\right]+2u^2\geq 0$$

$$g^2\left(t-\frac{3u\sin\theta}{2g}\right)^2-\frac{9}{4}u^2\sin^2\theta+2u^2\geq 0$$

The first term is a squared quantity, so it is always ≥ 0. The lowest possible value for the entire expression occurs when that squared term is exactly zero. For the overall expression to be non-negative at all times, its minimum value must be

$$\geq 0:\ 2u^2-\frac{9}{4}u^2\sin^2\theta\geq 0$$

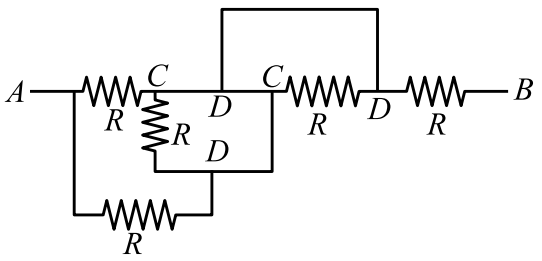
$$2\geq\frac{9}{4}\sin^2\theta\Rightarrow\frac{8}{9}\geq\sin^2\theta\Rightarrow\frac{2\sqrt{2}}{3}\geq\sin\theta$$

Hence option (2) correct.

(NCERT Ed. 2025-26, 11th, Page No. 39)

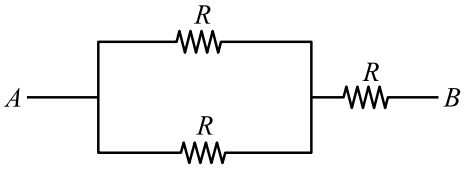
2. (1)

Let's label the junctions as:



Since there is a plain wire connecting C and D, this shorts the middle horizontal resistor.

Similarly, the node at the bottom of the vertical resistor is also connected to D, this shorts the vertical resistor as well.



$$R_{eq}=(R\,||\,R)+R=\frac{R}{2}+R=\frac{3R}{2}$$

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3. (2)

$$\vec{F}=-\left(\frac{\partial U}{\partial x}\hat{i}+\frac{\partial U}{\partial y}\hat{j}\right).\ \frac{\partial U}{\partial x}=2ax,\ \frac{\partial U}{\partial y}=4ay.$$

$$\therefore\vec{F}=-\left(2ax\hat{i}+4ay\hat{j}\right)=-2a\left(x\hat{i}+2y\hat{j}\right)$$

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4. (2)

At equilibrium: $f_{max}=mg\Rightarrow\mu F_m=mg$.

When $F=\frac{F_m}{5}$, friction $f=\mu\left(\frac{F_m}{5}\right)=\frac{mg}{5}$.

Net force $ma=mg-f=mg-\frac{mg}{5}=\frac{4mg}{5}$.

$$a=\frac{4g}{5}=\frac{4\times 10}{5}=8\ \text{m/s}^2$$

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5. (1)

Least Count (LC):

$$\text{Pitch}=\frac{\text{Distance moved}}{\text{No. of rotations}}=\frac{1\ \text{mm}}{2}=0.5\ \text{mm}$$

$$LC=\frac{\text{Pitch}}{\text{Total circular divisions}}$$

$$=\frac{0.5\ \text{mm}}{50}=0.01\ \text{mm}$$

Observed Reading:

$$\text{Observed}=\text{MSR}+(\text{CSR}\times LC)$$

$$=3+(25\times 0.01)=3.25\ \text{mm}$$

Applying Zero Correction:

$$\text{True Reading}=\text{Observed Reading}-(\text{Zero Error})$$

$$\text{True}=3.25-(-0.05)=3.30\ \text{mm}$$

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6. (2)

Max induced EMF $e = \frac{1}{2}BL^2\omega_{max}$.

At the bottom, $v_{max} = \sqrt{2gh}$

where $h = L(1-\cos\theta) = 2L\sin^2\left(\frac{\theta}{2}\right)$.

$v_{max} = \sqrt{2g\left(2L\sin^2\frac{\theta}{2}\right)} = 2\sin\left(\frac{\theta}{2}\right)\sqrt{gL}$.

Since $\omega_{max} = \frac{v_{max}}{L}$, then

$e = \frac{1}{2}BL^2\left(\frac{v_{max}}{L}\right) = \frac{1}{2}BLv_{max}$.

$e = \frac{1}{2}BL\left[2\sin\left(\frac{\theta}{2}\right)\sqrt{gL}\right] = BL\sin\left(\frac{\theta}{2}\right)\sqrt{gL}$.

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7. (2)

$Q_{\max} = C \times V_{\max} = C \times (E_b \times d)$

$= \frac{K\varepsilon_0 A}{d} \times E_b \times d = K\varepsilon_0 AE_b$

$Q_{\max} \propto KAE_b$

A: $3 \times 2 \times 2 = 12$

B: $4 \times 1 \times 4 = 16$

C: $2 \times 4 \times 1 = 8$

D: $10 \times 0.5 \times 2 = 10$

Capacitor B has the highest value.

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8. (3)

Power $P = (\text{Pressure}) \times (\text{Flow Rate})$.

Pressure $= \rho gh = 13600 \times 10 \times 0.1 = 13600$ Pa.

Flow $= \frac{6 \times 10^{-3}\ \text{m}^3}{60\ \text{s}} = 10^{-4}\ \text{m}^3/\text{s}$.

$P = 13600 \times 10^{-4} = 1.36$ W.

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9. (3)

The net voltage across inductor and capacitor is

$V_{LC} = |V_L - V_C| = |60 - 100| = 40$ V

Total source voltage

$V = \sqrt{V_R^2 + (V_L - V_C)^2}$

$\Rightarrow 50 = \sqrt{V_R^2 + 40^2} \Rightarrow V_R = 30$ V

Current $I = \frac{V_R}{R} = \frac{30}{10} = 3$ A

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10. (1)

Using $\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \Rightarrow d = \sqrt{\frac{1}{\sqrt{2}\pi n\lambda}}$

$d = \sqrt{\frac{1}{\sqrt{2}\pi \cdot 10^{14} \cdot \frac{10^{-6}}{\sqrt{2}\pi}}} = \sqrt{10^{-8}} = 10^{-4}$ cm.

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11. (4)

Diamagnetic materials develop induced dipoles in a direction opposite to the external field, resulting in a weak repulsive force regardless of the pole.

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12. (1)

Objective lens: Using $\frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o}$

$\frac{1}{v_o} = \frac{1}{8} - \frac{1}{9} = \frac{1}{72} \Rightarrow v_o = 72$ mm $= 7.2$ cm

Magnifying power (m):

$m = \frac{v_o}{|u_o|}\left(1 + \frac{D}{f_e}\right) = \frac{72}{9}\left(1 + \frac{25}{2.5}\right) = 8 \times 11 = 88$

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13. (4)

The magnetic force on a curved wire is

$F = i(\vec{L}_{eff} \times \vec{B})$.

For a semicircle, the effective length (displacement from A to C) is the diameter, $L_{eff} = 2R$.

$F = i(2R)B_0 = 2iRB_0$.

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14. (2)

Using $I' = I_{\max}\cos^2(\pi\Delta x/\lambda)$:

For $\Delta x = \lambda/3$, phase difference $\phi = 2\pi/3$.

$I' = I\cos^2(\pi/3) = I \times (1/2)^2 = \frac{I}{4}$.

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15. (1)

Net force on one particle towards center:

$F_{net} = \frac{Gm^2}{(2r)^2} + 2\frac{Gm^2}{(\sqrt{2}r)^2}\cos 45^\circ = \frac{Gm^2}{r^2}\left(\frac{1}{4} + \frac{1}{\sqrt{2}}\right)$.

Centripetal force $m\omega^2 r = \frac{Gm^2}{r^2}\left(\frac{1+2\sqrt{2}}{4}\right)$.

$\omega = \sqrt{\frac{Gm}{r^3}\left(\frac{1+2\sqrt{2}}{4}\right)}$

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16. (2)

Since $\lambda \propto 1/\Delta E$:

$\Delta E_{3\to1} = 2E - E = E \to \lambda$

$\Delta E_{2\to1} = 1.25E - E = 0.25E \to \dfrac{\lambda}{0.25} = 4\lambda$.

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17. (4)

Velocity of approach $= 10 - (-5) = 15$ m/s.

For elastic collision with heavy platform, velocity of separation $=$ approach $= 15$ m/s.

Final velocity of ball $v_f = 15 + 5 = 20$ m/s (upward).

Impulse

$= \Delta p = m(v_f - v_i) = 1[20 - (-10)] = 30$ N-s

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18. (4)

Resonant Frequencies:

Closed Pipe: $f_c = (2n-1)\dfrac{v}{4L}$ (where $n = 1, 2, 3, \ldots$)

Fundamental frequency $f_{0c} = \dfrac{340}{4\times2} = 42.5$ Hz.

Possible frequencies:

$42.5, 127.5, 212.5, 297.5, 382.5, 467.5, 552.5$ Hz,… (odd multiples).

Open Pipe: $f_o = m\dfrac{v}{2L}$ (where $m = 1, 2, 3, \ldots$)

Fundamental frequency $f_{0o} = \dfrac{340}{2\times2} = 85$ Hz.

Possible frequencies:

$85, 170, 255, 340, 425, 510$ Hz,… (all multiples).

Condition for Common Resonance: For a common frequency, we need: $(2n-1)\times 42.5 = m\times 85$

$(2n-1) = 2m$ Since $(2n-1)$ is always odd and $2m$ is always even, there is no frequency at which both pipes of equal length can resonate simultaneously.

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19. (2)

Magnitude: $B = \dfrac{E}{c} = \dfrac{12}{3\times10^8} = 4\times10^{-8}$ T.

Direction: $\vec{E}\times\vec{B}$ points in the direction of propagation ($+z$).

Since $\vec{E}$ is $+x$ ($\hat{i}$), $\vec{B}$ must be along $+y$ ($\hat{j}$) because $\hat{i}\times\hat{j} = \hat{k}$.

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20. (4)

Frequency $f = \dfrac{qB}{2\pi m} = \dfrac{(e/m)B}{2\pi}$.

$f = \dfrac{1.76\times10^{11}\times3.57\times10^{-3}}{2\times3.14} \approx \dfrac{6.28\times10^8}{6.28}$

$= 10^8$ Hz $= 100$ MHz

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21. (2)

The time period for a spring-mass system is $T = 2\pi\sqrt{\dfrac{m}{k}}$, which implies $k \propto \dfrac{1}{T^2}$.

For spring 1: $k_1 = \dfrac{4\pi^2 m}{t_1^2}$

For spring 2: $k_2 = \dfrac{4\pi^2 m}{t_2^2}$

Parallel Combination: When springs are in parallel, the equivalent stiffness is $k_{eq} = k_1 + k_2$.

Substituting the values: $\dfrac{4\pi^2 m}{t_0^2} = \dfrac{4\pi^2 m}{t_1^2} + \dfrac{4\pi^2 m}{t_2^2}$

$\Rightarrow \dfrac{1}{t_0^2} = \dfrac{1}{t_1^2} + \dfrac{1}{t_2^2}$ or $t_0 = \dfrac{t_1 t_2}{\sqrt{t_1^2 + t_2^2}}$

$t_0 = \dfrac{3\times4}{\sqrt{3^2+4^2}} = \dfrac{12}{\sqrt{25}} = \dfrac{12}{5} = 2.4$ s

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22. (1)

Decrease by 75% means $g' = 25\%$ of $g = \dfrac{g}{4}$.

Using $g' = g\left(\dfrac{R}{R+h}\right)^2 \Rightarrow \dfrac{1}{4} = \left(\dfrac{R}{R+h}\right)^2$

Taking square root:

$\dfrac{1}{2} = \dfrac{R}{R+h} \Rightarrow R + h = 2R \Rightarrow h = R$

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23. (4)

Rewrite the equation using $\sin^2\theta = \frac{1-\cos 2\theta}{2}$:

$x = x_0\left(\frac{1-\cos 2\omega t}{2}\right) = \frac{x_0}{2} - \frac{x_0}{2}\cos(2\omega t)$.

This represents SHM about the mean position $x = \frac{x_0}{2}$ with angular frequency $\omega' = 2\omega$.

Time Period $T = \frac{2\pi}{\omega'} = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$.

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24. (4)

Using $h = \frac{2T\cos\theta}{r\rho g} \Rightarrow T \propto \frac{h\rho}{\cos\theta}$

$\frac{T_w}{T_L} = \frac{h_w\rho_w\cos\theta_L}{h_L\rho_L\cos\theta_w} = \frac{6\times 1\times\left|\cos 120^\circ\right|}{2\times 9\times\cos 0^\circ}$

$= \frac{6\times 0.5}{18\times 1} = \frac{3}{18} = 1:6$

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25. (3)

By Boyle's Law:

$P_{\text{bottom}}V = P_{\text{surface}}(4V) \Rightarrow P_{\text{atm}} + h\rho_w g = 4P_{\text{atm}}$

$h\rho_w g = 3P_{\text{atm}} = 3(H\rho_m g) \Rightarrow h = 3H\left(\frac{\rho_m}{\rho_w}\right)$

$h = 3\times(0.7\text{ m})\times 10 = 21\text{ m}$

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26. (3)

To retrace the path, the ray must strike the silvered surface normally ($r_2 = 0$). Thus, $r_1 = A = 30^\circ$.

Using Snell's Law:

$\mu = \frac{\sin i}{\sin r_1} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\sqrt{3}/2}{1/2} = \sqrt{3} \approx 1.732$.

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27. (3)

Equivalent Conductivity (K_{eq}): For rods in series,

$K_{eq} = \frac{2K_1K_2}{K_1+K_2} = \frac{2\times 400\times 100}{400+100} = 160 \rightarrow$ (III)

Junction Temperature (T_j): Using continuity of heat current (H):

$400(T_j - 0) = 100(100 - T_j)$

$\Rightarrow 4T_j = 100 - T_j \Rightarrow T_j = 20 \rightarrow$ (IV)

Resistance Ratio: Since $R = \frac{L}{KA}$ and L, A are same, $\frac{R_{Cu}}{R_{St}} = \frac{K_{St}}{K_{Cu}} = \frac{100}{400} = 0.25 \rightarrow$ (I)

Heat Flow Rate (H):

$H = \frac{K_{eq}\cdot A\cdot\Delta T}{2L} = \frac{160\times 0.1\times 100}{2} = 800 \rightarrow$ (II)

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28. (1)

The ionization energy decreases as the distance of bonding electrons from the nucleus increases. In Carbon ($n = 2$), the electrons are more tightly bound, leading to a much larger energy gap ($E_g \approx 5.4$ eV) compared to Silicon ($n = 3$, $E_g \approx 1.1$ eV). Thus, C is an insulator and Si is a semiconductor.

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29. (4)

According to Stefan-Boltzmann Law, the rate of loss of heat $\left(\frac{dQ}{dt}\right)$ by radiation for a body at temperature T in surroundings at temperature T_0 is given by: $\frac{dQ}{dt} = \sigma eA(T^4 - T_0^4)$ Where:

Since material, surface finish, and temperature are the same, the rate of loss of heat depends only on the surface area (A): $\frac{dQ}{dt} \propto A$

For a sphere, $A = \pi d^2$, so: $\frac{dQ}{dt} \propto d^2$

Given the ratio of diameters $d_1 : d_2 = 1:2$, the ratio of the rate of loss of heat is:

$\frac{(dQ/dt)_1}{(dQ/dt)_2} = \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

Therefore, $(dQ/dt)_2 = 4\times(dQ/dt)_1$.

The initial rate of loss of heat for the larger ball is four times that of the smaller ball.

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30. (3)

Einstein's equation $K_{max} = h\nu - \phi$ gives the maximum kinetic energy. Electrons from deeper levels or those undergoing collisions lose more energy and emerge with $K < K_{max}$.

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31. (3)

Elastic Potential Energy $U = \dfrac{F^2 L}{2AY}$ where $A = \pi r^2$.

Values: $r = 10^{-2}$ m $\Rightarrow A = \pi \times 10^{-4}$ m^2,

$F = 4 \times 10^4$ N, $L = 2\pi$ m.

$$U = \frac{(4\times10^4)^2 \cdot 2\pi}{2\cdot(\pi\times10^{-4})\cdot 2\times10^{11}} = \frac{32\pi\times10^8}{4\pi\times10^7} = 80 \text{ J}$$

(NCERT Ed. 2025-26, 11th, Page No. 174)

32. (2)

I is Incorrect: Normal reaction $N = mg - F_2$.

As F_2 increases, N decreases.

II is Correct: Since the block does not slide, static friction f_s equals the applied horizontal force F_1. Since F_1 is constant, f_s remains constant.

III is Correct: Limiting friction $f_L = \mu_s N$. Since N decreases as F_2 increases, f_L decreases.

(NCERT Ed. 2025-26, 11th, Page No. 60)

33. (4)

At Highest Point:

Horizontal distance (Half Range):

$$x = \frac{R}{2} = \frac{u^2 \sin\theta\cos\theta}{g}$$

Vertical height: $H = \dfrac{u^2\sin^2\theta}{2g}$

Velocity: $v = u\cos\theta$ (horizontal only)

Torque (τ): Torque is due to the force of gravity (mg) acting at a horizontal lever arm x.

$$\tau = \text{Force} \times \text{Perpendicular distance} = (mg)\cdot\frac{R}{2}$$

$$= mg\left(\frac{u^2\sin\theta\cos\theta}{g}\right) = mu^2\sin\theta\cos\theta$$

Angular Momentum (L): Angular momentum is the product of linear momentum and the perpendicular distance to the velocity vector (which is the height H).

$$L = \text{Momentum} \times \text{Height} = (mu\cos\theta)\cdot H$$

$$= mu\cos\theta\left(\frac{u^2\sin^2\theta}{2g}\right) = \frac{mu^3\sin^2\theta\cos\theta}{2g}$$

Ratio ($\dfrac{\tau}{L}$):

$$\text{Ratio} = \frac{mu^2\sin\theta\cos\theta}{\left(\dfrac{mu^3\sin^2\theta\cos\theta}{2g}\right)} = \frac{1}{\left(\dfrac{u\sin\theta}{2g}\right)} = \frac{2g}{u\sin\theta}$$

(NCERT Ed. 2025-26, 11th, Page No. 106)

34. (1)

Energy released (Q) in a nuclear reaction is the difference between the total binding energy of the products and the total binding energy of the reactants.

Reaction: $3B \rightarrow 1A + Q$

$Q = (\text{B.E. of Product}) - (\text{B.E. of Reactants})$

$= E_a - 3E_b$.

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35. (1)

According to the principle of homogeneity, each term in the sum must have the same dimension as y ($[L]$): $[a] = [y] = L$

$[bt] = L \Rightarrow [b] = LT^{-1}$

$[ct^2] = L \Rightarrow [c] = LT^{-2}$

The dimensions of the requested ratio:

$$\text{Ratio} = \frac{[a][c]}{[b]} = \frac{(L)\cdot(LT^{-2})}{LT^{-1}} = \frac{L^2T^{-2}}{LT^{-1}} = LT^{-1}.$$

Since $[y] = L$ and $[t] = T$, then LT^{-1} is the same as the dimension of $\dfrac{y}{t}$.

(NCERT Ed. 2025-26, 11th, Page No. 33)

36. (2)

Temperatures in Kelvin: $T_1 = 500$ K, $T_2 = 400$ K

Efficiency $\eta = 1 - \dfrac{T_2}{T_1} = 1 - \dfrac{400}{500} = \dfrac{1}{5}$

Work done $W = \eta Q_1 = \dfrac{1}{5}\times 12 \text{ kJ} = 2.4 \text{ kJ}$

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[Contd...

37. (2)

Gate I (AND): $0 \cdot 1 = 0$

Gate II (NOR): $\overline{0+1} = \overline{1} = 0$

Gate III (NAND): $\overline{1 \cdot 1} = \overline{1} = 0$

Gate IV (OR): $0+1=1$

Only Gate IV has an output of 1; Gates I, II, and III have an output of 0.

38. (4)

Circumference $S = 2\pi R = 2\pi(0.5) = \pi$ m

Relative speed $v_{\text{rel}} = 25\pi - 20\pi = 5\pi$ cm/s

Time to meet $t = \dfrac{S}{v_{\text{rel}}} = \dfrac{100\pi}{5\pi} = 20$ s

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39. (2)

Centre of Mass Position ($\vec{r}_{cm}$):

$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2} = \frac{1\left(2\hat{i}+5\hat{j}+3\hat{k}\right)+2\left(2\hat{i}-\hat{j}\right)}{3}$$

$$= \frac{6\hat{i}+3\hat{j}+3\hat{k}}{3} = 2\hat{i}+\hat{j}+\hat{k}$$

Magnitude:

$$|\vec{r}_{cm}| = \sqrt{2^2+1^2+1^2} = \sqrt{6}$$

Comparison:

The vector in option (2), $\left(\hat{i}+2\hat{j}+\hat{k}\right)$, has a magnitude of $\sqrt{1^2+2^2+1^2} = \sqrt{6}$.

Therefore, its magnitude is identical.

(NCERT Ed. 2025-26, 11th, Page No. 96)

40. (3)

A conductor in an electrostatic field is an equipotential volume. Therefore, the potential at every point on the surface and inside the hollow conducting sphere is the same.

(NCERT Ed. 2025-26, 12th, Page No. 62)

41. (1)

An arrangement of n plates with alternate connections forms $(n-1)$ identical capacitors connected in parallel.

$$C_{eq} = (n-1) \times C_{single} = \frac{(n-1)\epsilon_0 A}{d}$$

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42. (1)

Distance is the area under the $v-t$ graph. Area $OC = \dfrac{1}{2} \times OC \times h$; Area $CB = \dfrac{1}{2} \times CB \times h$.

Ratio $= \dfrac{OC}{CB}$.

From trigonometry: $OC = \dfrac{h}{\tan 30^\circ} = h\sqrt{3}$ and

$$CB = \frac{h}{\tan 60^\circ} = \frac{h}{\sqrt{3}}.$$

Ratio $= \dfrac{h\sqrt{3}}{h/\sqrt{3}} = \dfrac{3}{1} = 3:1$

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43. (2)

Using Gauss's Law, $q = \epsilon_0 \Phi = \epsilon_0 (E \cdot A)$

At $r = 0.2$ m, $E = 500 \times 0.2 = 100$ V/m

Area $A = 4\pi(0.2)^2 = 0.16\pi$ m^2

$q = \epsilon_0 (100 \times 0.16\pi) = 16\pi\epsilon_0$ C

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44. (3)

For field to be zero,

$$\frac{kQ}{x^2} = \frac{k(4Q)}{(L-x)^2} \Rightarrow \frac{1}{x} = \frac{2}{L-x} \Rightarrow L - x = 2x.$$

Solving gives $x = \dfrac{L}{3}$

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45. (2)

Resistance $R = \dfrac{V^2}{P}$, so lower wattage means higher resistance. In series, $V = IR$; thus, the lamp with higher resistance (40W) has the larger potential drop.

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[Contd...

46. (2)

$$n(\text{dibenzalacetone}) = \frac{585}{234} = 2.5 \text{ mol}$$

From stoichiometry:
1 mol dibenzalacetone ⇒ 2 mol benzaldehyde
So, n(benzaldehyde) = 2 × 2.5 = 5.0 m

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

47. (2)

For h-subshell, $\ell = 5$, so number of orbitals equals $2\ell + 1 = 11$.

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48. (3)

$3d_{z^2}$ has electron density in XY plane, while A, C, and D are correct.

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49. (3)

Endothermic needs positive ΔH and spontaneity needs ΔG negative, satisfied when TΔS exceeds ΔH.

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50. (1)

From above, work function: $\phi = 2.10$ eV

Threshold wavelength: $\lambda_0 = \frac{hc}{\phi}$

Using $hc = 12420$ eV Å,

$$\lambda_0 = \frac{12420}{2.10} \approx 5946 \text{ Å}$$

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51. (1)

By first law, ΔU = q − w = 30 − 12 = 18 J.

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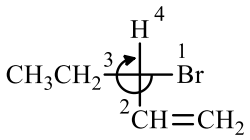
52. (3)

$$pH = 7 + \frac{1}{2}(5.23 - 4.75) = 7.24$$

Since pH > 7, the solution is slightly basic.

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53. (1)

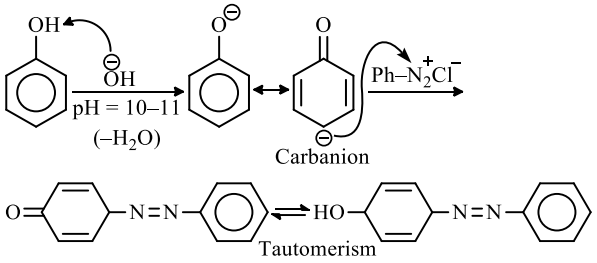


(NCERT Ed. 2025-26 12th Part-II Page No. 176)

54. (2)

Cu^{2+} forms black CuS in acidic medium, while zinc sulphide precipitates in alkaline medium.

55. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 276)

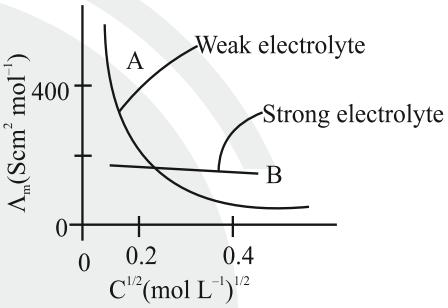
56. (2)

Five grams per 100 g solution means

$$\frac{1000 \times 5 \times 1.2}{100} = 60 \text{ g L}^{-1}.$$

(NCERT Ed. 2025-26 12th Part-I Page No. 2)

57. (3)



(NCERT Ed. 2025-26 12th Part-I Page No. 47)

58. (3)

$$\Delta H - \Delta U = \Delta n_g RT$$

$\Delta n_g = 1, T = 300$ K

$$\Delta H - \Delta U = \frac{8.314 \times 300}{1000} = 2.49 \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 161)

59. (2)

First-order half-life equals 0.693/k, so it is independent of initial concentration.

(NCERT Ed. 2025-26 12th Part-I Page No. 76)

60. (3)

In both complexes iron is Fe^{2+}, so it is $3d^6$. CN^- is a strong-field ligand and causes pairing, giving a low-spin octahedral complex with no unpaired electron, whereas F^- is weak field and gives four unpaired electrons.

(NCERT Ed. 2025-26 12th Part-I Page No. 129)

61. (2)

$$C_6H_5OCH_3 + HI \xrightarrow{\Delta} C_6H_5OH + CH_3I$$

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62. (1)

[Contd...

$n_X = 0.03, \quad n_Y = 0.01$

Mole fraction of Y: $x_Y = \frac{0.01}{0.04} = 0.25$

$P_Y = 0.25 \times 740 = 185$ mm Hg

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63. (4)

Europium belongs to lanthanoids, whereas neptunium, curium and thorium are actinoids.

(NCERT Ed. 2025-26 12th Part-I Page No. 111)

64. (3)

By definition, an ideal solution obeys Raoult's law at all compositions.

(NCERT Ed. 2025-26 12th Part-I Page No. 13)

65. (3)

Rate $= K[A]^X[B]^Y$

$6.0 \times 10^{-3} = K[0.1]^y[0.1]^y$

$2.4 \times 10^{-2} = K[0.4]^x[0.1]^y$

x = 1

$7.2 \times 10^{-2} = K[0.3]^x[0.2]^y$

$2.88 \times 10^{-1} = K[0.3]^x[0.4]^y$

y = 2

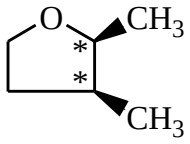
(NCERT Ed. 2025-26 12th Part-I Page No. 67)

66. (2)

Hydrogen electrode and silver-silver bromide electrode together represent exactly the given overall cell reaction.

(NCERT Ed. 2025-26 12th Part-I Page No. 55)

67. (2)



2 Chiral centers

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68. (1)

$n(\text{pent-1-ene}) = \frac{5.0}{70} = 0.0714$ mol

Since pent-1-ene and Br_2 react in 1:1 ratio:

$n(Br_2) = 0.0714$ mol

(NCERT Ed. 2025-26 11th Part-I Page No. 20)

69. (3)

Photoelectric emission is instantaneous because one photon interacts with one electron and directly gives sufficient energy when frequency exceeds threshold, so both statements are correct and the reason explains the assertion.

(NCERT Ed. 2025-26 11th Part-I Page No. 41)

70. (2)

NH_3 and CO_2 satisfy the octet rule for the central atom, whereas BCl_3 has six electrons, $BeCl_2$ has four, and PF_5 has an expanded octet, so three species do not have eight electrons.

(NCERT Ed. 2025-26 11th Part-I Page No. 101)

71. (2)

Assume 100 g of compound, so carbon is 80 g and hydrogen is 20 g. Moles are $\frac{80}{12}$ and 20 respectively. Dividing by the smaller value gives C:H = 1:3, hence the empirical formula is CH_3.

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72. (1)

In multielectron atoms, energy depends on both n and *l*, but not on m_l when external fields are absent. Thus the two orbitals having same n and *l* are degenerate.

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73. (1)

$\frac{E_a}{2.303\,RT} = \frac{26000}{T}$

$E_a = 26000 \times 2.303 \times 8.314 = 498$ kJ

(NCERT Ed. 2025-26 12th Part-I Page No. 79)

74. (1)

$G = \frac{1}{R} = \frac{1}{1750} = 5.71 \times 10^{-4}$ S

$GG^* = \kappa$

$\frac{0.152 \times 10^{-3}}{5.71 \times 10^{-4}} = \frac{\ell}{a}; \qquad \frac{\ell}{a} = 2.66 \times 10^{-1}$

(NCERT Ed. 2025-26 12th Part-I Page No. 46)

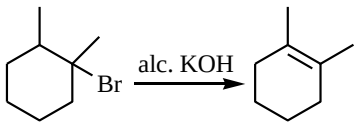
75. (2)

Using $\Delta G^o = -nF E^o_{cell}$

So, $n E^o_{cell} = -\frac{\Delta G^o}{F} = -0.18$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

76. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 180)

77. (2)

[Contd...

Neon has a highly positive electron gain enthalpy because of its stable noble gas configuration, whereas chlorine has a highly negative value due to strong tendency to gain one electron. Hence their difference is maximum.

78. (1)

In cyclobutane, dichloro substitution can give 1,1-, 1,2- and 1,3- derivatives by symmetry. Considering positional isomers only, these are the only distinct dichloro derivatives possible.

(NCERT Ed. 2025-26 12th Part-II Page No. 165)

79. (3)

Second-period elements cannot expand octet, so N, O and F fail, while B also does not show valency 3? Thus N, O, F and C.

80. (2)

Lone pair of nitrogen involved in conjugation, so hybridisation of N is sp^2.

(NCERT Ed. 2025-26 11th Part-I Page No. 115)

81. (3)

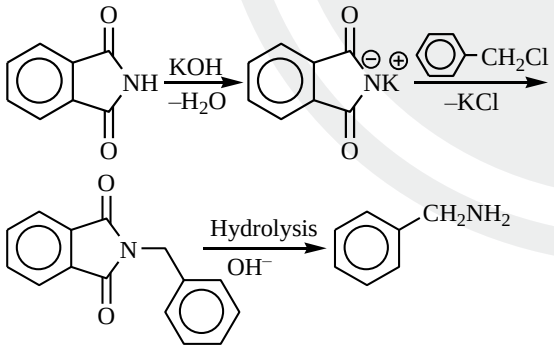
Oxidation number of S in $S_8 = 0$

Oxidation number of S in S_2F_2: +1

Oxidation number of S in H_2S: –2

(NCERT Ed. 2025-26 11th Part-II Page No. 240)

82. (4)



(NCERT Ed. 2025-26 12th Part-II Page No. 264)

83. (2)

Moles of glucose: $n_g = \frac{3.69}{180} = 0.0205$

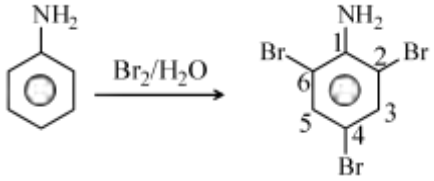
Moles of water: $n_w = \frac{100}{18} = 5.56$

For dilute solution: $\frac{\Delta P}{P^0} = \frac{n_g}{n_w}$

$$\Delta P = 54.2 \times \frac{0.0205}{5.56} = 0.20 \text{ mm Hg}$$

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84. (4)



2,4,6 -Tribromo aniline
(white ppt)

(NCERT Ed. 2025-26 12th Part-II Page No. 272)

85. (1)

The example of partition chromatography is paper chromatography.

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86. (1)

- In chromatography, silica gel is used as absorbent.
- In Dumas method, N_2 gas is produced during the process.
- In Kjeldahl's method, nitrogen in the organic compound is converted into ammonium sulphate.
- In Carius method, organic compound is heated with fuming nitric acid in the presence of $AgNO_3$.

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87. (1)

Reduction half reaction:

$$MnO_4^- + 5e^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O] \times 2$$

$$\text{Oxidation half reaction}: C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-] \times 5$$

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

(NCERT Ed. 2025-26 11th Part-II Page No. 248)

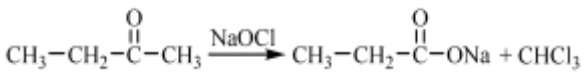
88. (2)

Potash alum has potassium sulphate and aluminium sulphate with 24 molecules of water of crystallization. i.e. $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(NCERT Ed. 2025-26 12th Part-I Page No. 120)

89. (1)

Haloform reaction:



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90. (1)

Alcohols and phenols can be distinguished by neutral $FeCl_3$. Phenols give violet coloured complex while alcohols does not react with neutral $FeCl_3$.

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[Contd...

91. (3)

For a linear DNA, if there are n cut sites total, complete digestion gives n + 1 fragments.

Here:

- Enzyme X has 3 recognition sites
- Enzyme Y has 2 recognition sites

So total cut sites = 3 + 2 = 5

Hence, maximum number of fragments = 5 + 1 = 6

Why fragment visibility depends on molecular weight:

On an agarose gel, DNA fragments are separated primarily on the basis of their size/molecular weight. Smaller fragments move faster and farther than larger ones. So the pattern you observe depends on the length of the fragments produced.

(NCERT Ed. 2025-26, 12th, Page No. 168)

92. (4)

Haplodiploid sex determination is a system where males develop from unfertilized eggs and are haploid (n), while females develop from fertilized eggs and are diploid (2n). This system is most famously seen in insects like honey bees.

Three pairs of jointed legs present: This describes an insect. Since haplodiploidy is very common in insects (like honeybees), this can be true.

Production of gametes in males by mitosis: Since males are already haploid (n), they cannot undergo meiosis to produce gametes. Therefore, they produce sperm via mitosis. This is true.

Females have double the number of chromosomes as compared to males: In this system, females are 2n and males are n. Therefore, females have exactly twice the number of chromosomes. This is true.

Placed in phylum Chordata: Haplodiploidy is characteristic of certain Arthropods. It is not found in the Phylum Chordata (which includes vertebrates like humans, birds, and fish), where sex is typically determined by specific sex chromosomes (like XY or ZW) or environment. Therefore, this cannot be true.

(NCERT Ed. 2025-26, 12th, Page No. 71)

93. (2)

Marginal plants are producers that grow at the edges of water bodies. Primary consumers: are herbivores (animals that eat plants). The third trophic level is typically occupied by secondary consumers (carnivores that eat herbivores).

(NCERT Ed. 2025-26, 12th, Page No. 206)

94. (1)

The cell envelope in bacteria consists of a tightly bound three layered structure i.e., the outermost glycocalyx followed by the cell wall and then the plasma membrane. Although each layer of the envelope performs distinct function, they act together as a single protective unit. Mesosomes help in cell wall formation, DNA replication and distribution to daughter cells. They also help in respiration, secretion processes, to increase the surface area of the plasma membrane and enzymatic content. The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics. The fimbriae are small bristle like fibres sprouting out of the cell. In some bacteria, they are known to help attach the bacteria to rocks in streams and also to the host tissues.

(NCERT Ed. 2025-26, 11th, Page No. 90)

95. (4)

List-I (Animal)	List-II (Excretory waste)
Labeo	Ammonia
Carcharadon	Urea
Rana	Ammonia in larva and urea in adult
Calotes	Uric acid

(NCERT Ed. 2025-26, 11th, Page No. 205)

96. (3)

Blue-white screening is a technique used to identify cells that have successfully taken up a recombinant plasmid. In this process, non-recombinant colonies produce a functional enzyme that turns a substrate blue. However, when a foreign gene is inserted into the plasmid, it disrupts the enzyme's gene (insertional inactivation), so recombinant colonies remain white. Transformants are defined as any cells that have successfully picked up a plasmid, whether it contains the insert or not.

(NCERT Ed. 2025-26, 12th, Page No. 170)

97. (3)

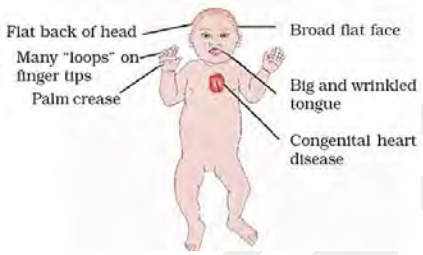
In *Oxalis*, stem is modified for spread and vegetative propagation. In pitcher plant (*Nepenthes*), leaf is modified into a pitcher. In *Rhizophora*, roots come out of the ground and grow vertically upwards. Such roots, called pneumatophores, help to get oxygen for respiration. In the parasitic plant *Cuscuta*, leaves have lost chlorophyll in the course of evolution.

(NCERT Ed. 2022-23, 11th, Page No. 68)

[Contd...

98. (1)

Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy. For example, Down's syndrome results in the gain of extra copy of chromosome 21. This disorder was first described by Langdon Down (1866). The affected individual is short statured with small round head, furrowed tongue and partially open mouth. Palm is broad with characteristic palm crease. Physical, psychomotor and mental development is retarded. This disorder can occur in both males as well as females.



A representative figure showing an individual inflicted with Down's syndrome

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99. (3)

In gymnosperms, the multicellular female gametophyte is the tissue that bears two or more archegonia (the female sex organs). The gametophyte itself is not "called" archegonia; rather, archegonia are structures contained within it. *Cedrus* is a gymnosperm. The word "gymnosperm" literally means "naked seeds" (gymnos: naked, sperma: seeds). Because their ovules are not enclosed within an ovary wall, they do not produce true fruits. The stems in gymnosperms are unbranched (*Cycas*) or branched (*Pinus, Cedrus*). Unlike bryophytes and pteridophytes, in gymnosperms, the male and the female gametophytes do not have an independent free-living existence.

(NCERT Ed. 2025-26, 11th, Page No. 32)

100. (4)

Plants have different kinds of meristems. The meristems which occur at the tips of roots and shoots and produce primary tissues are called apical meristems. Root apical meristem occupies the tip of a root while the shoot apical meristem occupies the distant most region of the stem axis. During the formation of leaves and elongation of stem, some cells 'left behind' from shoot apical meristem, constitute the axillary bud. Such buds are present in the axils of leaves and are capable of forming a branch or a flower. The meristem which occurs between mature tissues is known as intercalary meristem. They occur in grasses and regenerate parts removed by the grazing herbivores. Both apical meristems and intercalary meristems are primary meristems because they appear early in life of a plant and contribute to the formation of the primary plant body. The meristem that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis and appear later than primary meristem is called the secondary or lateral meristem. They are cylindrical meristems. Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of lateral meristems. These are responsible for producing the secondary tissues.

(NCERT Ed. 2022-23, 11th, Page No. 85)

101. (3)

In a large-scale industrial setup, a stirred-tank bioreactor is preferred over simple shake flasks because it allows for a continuous culture system. By providing optimal conditions such as controlled temperature, pH, and oxygen availability, it helps maintain the cells in their most active logarithmic or exponential growth phase. This ensures a much higher yield of the desired enzyme or protein compared to the limited capacity and nutrient exhaustion typically seen in shake flasks.

(NCERT Ed. 2025-26, 12th, Page No. 174)

102. (2)

List-I	List-II
Urochordata	Notochord present only in larval tail
Cephalochordata	Notochord persists throughout life from head to tail
Cyclostomata	Sucking and circular mouth without jaws
Osteichthyes	Skin is covered with cycloid/ ctenoid scales

(NCERT Ed. 2025-26, 11th, Page No. 46, 47, 48)

103. (3)

List-I	List-II
Tryptophan	Aromatic amino acid
Aspartic acid	Acidic amino acid
Curcumin	Drug
Ricin	Toxin

(NCERT Ed. 2025-26, 11th, Page No. 106, 108)

[Contd...

104. (3)

The cortical layer that has casparian strips in dicot roots is endodermis, which has barrel-shaped cells. The layer present on the inner side of endodermis in dicot stem is the pericycle. In a dicot stem, the pericycle is described as occurring in semi-lunar patches of sclerenchyma above the phloem. In a dorsiventral (dicot) leaf, the adaxial (upper) mesophyll is the palisade parenchyma. These cells are described as elongated and vertically arranged. It is the abaxially placed (lower) spongy parenchyma that consists of oval or round cells. The structures that regulate the process of transpiration and gaseous exchange in monocots is the guard cells of the stomata. In monocots (grasses), the guard cells are dumb-bell shaped. In dicots, they are bean-shaped.

(NCERT Ed. 2025-26, 11th, Page No. 77)

105. (3)

The common Indian frog, *Rana tigrina*, has specific adaptations for its amphibious lifestyle. While it uses its skin for respiration in water (cutaneous) and lungs on land (pulmonary), its fertilization is external, occurring in the water where eggs and sperm are released. It also undergoes aestivation (summer sleep) and hibernation (winter sleep) to survive extreme environmental conditions.

(NCERT Ed. 2025-26, 11th, Page No. 84)

106. (3)

The ampullary region of the fallopian tube (oviduct) is the specific site where fertilization—the fusion of the egg and sperm—normally takes place. If both oviducts are completely blocked in this region, the sperm will be unable to reach the egg. Consequently, even if the woman ovulates and has normal menstrual cycles, fertilization becomes highly unlikely due to the physical obstruction of the meeting point.

(NCERT Ed. 2025-26, 12th, Page No. 35)

107. (3)

The diagram showing two distinct peaks forming from a single original population illustrates disruptive selection. This occurs when environmental conditions favor individuals at both extremes of a trait over those with intermediate values. A classic example is a bird population where those with very small beaks (for small seeds) and very large beaks (for large seeds) survive better than those with medium-sized beaks, eventually splitting the population into two phenotypic groups.

(NCERT Ed. 2025-26, 12th, Page No. 121)

108. (2)

In human reproduction, specific terms describe the stages of gamete formation. Spermiogenesis is the actual transformation of circular spermatids into elongated, motile spermatozoa; the term "spermiation" refers only to their release from the Sertoli cells. In females, the process of oogenesis begins before birth, and all primary oocytes are suspended or "arrested" in the Prophase I stage of meiosis until the female reaches puberty.

(NCERT Ed. 2025-26, 12th, Page No. 31, 32)

109. (4)

During the menstrual cycle, if fertilization does not occur, the corpus luteum begins to degenerate. This leads to a sharp decline in progesterone levels, which is the hormone responsible for maintaining the endometrial lining of the uterus. Without progesterone, the lining breaks down and is shed through the vagina, resulting in the phase known as menstrual bleeding.

(NCERT Ed. 2025-26, 12th, Page No. 34)

110. (4)

In the digestive system of a cockroach, the crop is a large, sac-like structure used primarily for the storage of food. While some initial breakdown may occur, its main role is not digestion itself. The grinding of food into fine particles is the job of the gizzard (proventriculus), which contains powerful chitinous teeth. The foregut is indeed lined with cuticle, and hepatic caeca are located at the junction of the foregut and midgut to secrete digestive juices.

(NCERT Ed. 2020-21, 11th, Page No. 113)

111. (4)

The evolutionary history of vertebrates shows a clear transition from aquatic to terrestrial life. It began with the appearance of jawless fish, followed by the development of lobefin fish which could move on land. These lobefins evolved into the first amphibians, which eventually gave rise to reptiles. This sequence highlights the gradual adaptation of body systems to survive outside of a water environment.

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112. (3)

Morphine is a powerful analgesic (painkiller) and sedative that is widely used in medicine. It is a natural alkaloid extracted from the milky latex of the unripe seed capsules of the poppy plant, scientifically known as *Papaver somniferum*. While other plants like *Cannabis* or *Erythroxylum* (coca) produce different types of drugs, the poppy plant is the specific source for opiates like morphine and heroin.

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[Contd...

113. (4)

Standard diagnostic methods often fail to detect a pathogen until its numbers are high enough to cause symptoms. However, PCR (Polymerase Chain Reaction) is a powerful molecular tool that can amplify a tiny, almost undetectable amount of a pathogen's DNA or RNA into millions of copies. This allows doctors to identify infections like HIV or certain cancers at a very early stage, long before any physical symptoms appear.

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114. (3)

Feature A: Asymmetrical, Acoelomate

- Animals in this category lack a definite shape and do not have a body cavity.
- Match: Porifera (Sponges) are the only major phylum where most members are asymmetrical. Since they only have a cellular level of organization, they lack a true coelom.

Feature B: Radial Symmetry, Acoelomate

- These animals can be divided into identical halves by any plane passing through the central axis but lack a body cavity between the gut and the body wall.
- Match: Ctenophora (Comb jellies) and Coelenterata (Cnidarians) exhibit radial symmetry and are acoelomate.

Feature C: Bilateral Symmetry, Pseudocoelomate

- These animals have a body that can be divided into identical left and right halves. Their body cavity is not lined by mesoderm; instead, the mesoderm is present as scattered pouches.
- Match: Aschelminthes (Roundworms) are the unique, "textbook" example of pseudocoelomates.

Feature D: Bilateral Symmetry, Coelomate

- These are higher animals with a true body cavity completely lined by mesoderm.
- Match: Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata, and Chordata all fit this description.

(NCERT Ed. 2025-26, 11th, Page No. 38, 39, 40)

115. (3)

The work of Mendel and others who followed him gave us an idea of inheritance patterns. However the nature of those 'factors' which determine the phenotype was not very clear. As these 'factors' represent the genetic basis of inheritance, understanding the structure of genetic material and the structural basis of genotype and phenotype conversion became the focus of attention in biology for the next century. The entire body of molecular biology was a consequent development with major contributions from Watson, Crick, Nirenberg, Khorana, Kornbergs (father and son), Benzer, Monod, Brenner, etc.

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116. (3)

Cork cambium is responsible for secondary growth. Most monocots do not undergo secondary growth as they lack lateral meristems (vascular cambium and cork cambium). While bulliform cells are a characteristic feature of monocot leaves (especially grasses), they are located on the adaxial (upper) epidermis, not the abaxial (lower) epidermis. They help in leaf rolling to prevent water loss. In monocot roots, the pith is large and well-developed. In contrast, dicot roots have a very small or inconspicuous pith. One of the defining anatomical features of a monocot stem is that phloem parenchyma is absent.

(NCERT Ed. 2025-26, 11th, Page No. 74)

117. (2)

Humans have always depended on nature for food and shelter, but when 'need' turns to 'greed', it leads to over-exploitation of natural resources. Many species extinctions in the last 500 years (Steller's sea cow, passenger pigeon) were due to over-exploitation by humans.

(NCERT Ed. 2025-26, 12th, Page No. 222)

118. (3)

The human skeletal system is divided into the axial and appendicular portions. The axial skeleton (80 bones) includes the skull, vertebral column, and ribs. There are 12 pairs of ribs, and the first seven pairs are called "true ribs" because they are directly attached to the sternum ventrally and the vertebral column dorsally. The other options are incorrect because there are 8 cranial bones, and the appendicular skeleton (not the whole body) has 126 bones.

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119. (2)

Seed colour in pea (II - Dominance): In Mendel's experiments with garden pea, yellow seed colour is completely dominant over green seed colour.

Flower colour in snapdragon (IV - Incomplete dominance): In *Antirrhinum* sp., a cross between red-flowered and white-flowered plants produces pink-flowered F1 offspring, where neither allele is completely dominant over the other.

Skin colour in humans (I - Polygenic inheritance): Human skin colour is controlled by multiple genes (typically more than three), resulting in a wide range of phenotypes based on the number of dominant alleles.

Color blindness in humans (III - X-linked): Color blindness is a sex-linked recessive disorder where the defective gene is located on the X-chromosome.

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[Contd...

120. (4)

In prokaryotes, ribosomes are associated with the plasma membrane of the cell. They are about 15 nm by 20 nm in size and are made of two subunits - 50S and 30S units which when present together form 70S prokaryotic ribosomes.

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121. (2)

In cancer biology, a critical distinction is made between tumor types. Benign tumors stay localized and do not spread, whereas malignant tumors are composed of neoplastic cells that grow rapidly and invade surrounding tissues. A defining characteristic of malignant tumors is metastasis, the ability of cells to slough off and reach distant sites through the blood to form new secondary tumors. Cancer cells also lose the property of contact inhibition, which normally stops cell growth when cells touch each other.

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122. (4)

The interaction where one species is benefitted and the other is neither benefitted nor harmed is called commensalism.

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123. (2)

Scientist		**Experimental material**
T. H Morgan	-	Fruitfly
Frederick Griffith	-	*Streptococcus pneumoniae*
Jacob and Monod	-	*E. coli*
Taylor and colleagues	-	*Vicia faba*

(NCERT Ed. 2025-26, 12th, Page No. 84)

124. (2)

The Polymerase Chain Reaction (PCR) is a three-step cycle. First, high heat is used to "denature" or separate the two strands of DNA. Second, the mixture is cooled so that short, single-stranded DNA primers can "anneal" or bind to the specific target regions. Finally, a heat-stable enzyme called Taq polymerase extends the primers to create new DNA strands. By repeating this cycle 30 times, a single piece of DNA can be amplified into over a billion copies.

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125. (3)

Eosinophils are a type of granulocyte in the blood that play a crucial role in the body's immune response. They are primarily associated with resisting infections and are specifically involved in allergic reactions and defense against parasitic infections. While other cells like neutrophils are phagocytic, eosinophils are best known for their increased count during allergies and their ability to release chemicals that combat parasites.

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126. (2)

In biotechnology, "competence" refers to the ability of a bacterial cell to take up foreign DNA from its surroundings. To make *E. coli* cells competent, they are typically treated with a specific concentration of a divalent cation, such as calcium (Ca^{2+}). This is followed by a "heat shock" procedure where the cells and DNA are briefly incubated at 42°C, which creates temporary pores in the bacterial cell wall through which the DNA can enter.

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127. (1)

The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth. Ecologists and evolutionary biologists have proposed various hypotheses that might account for greater biological diversity in tropics. Some important ones are (a) Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification, (b) Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity and (c) There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

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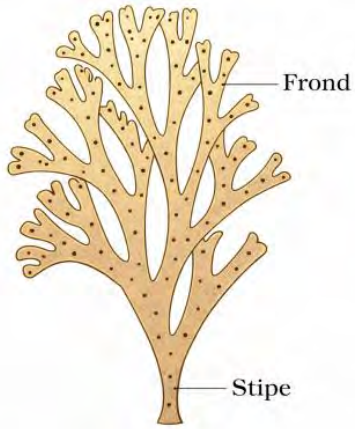
128. (4)

There is a characteristic distribution of the cells within the embryo sac. Three cells are grouped together at the micropylar end and constitute the egg apparatus. The egg apparatus, in turn, consists of two synergids and one egg cell. The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid. Three cells are at the chalazal end and are called the antipodals. The large central cell has two polar nuclei. Thus, a typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

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129. (4)

The organism shown in the image is the brown alga *Dictyota*. It does not possess air bladders. Air bladders are characteristic of other brown algae like *Fucus* or *Sargassum*. In this diagram, 'X' points to the frond (the leaf-like photosynthetic organ). The structure 'Y' is narrowed, stalk-like region called the stipe. The holdfast is the flattened, disc-like structure at the very bottom used for attachment. In brown algae, asexual reproduction occurs by motile zoospores that are pear-shaped (pyriform) and have two unequal, laterally attached flagella. Sexual reproduction (whether isogamous, anisogamous, or oogamous) involves gametes that are typically pyriform (pear-shaped) and bear two laterally attached flagella.



Dictyota

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130. (1)

Auxins like IAA and indole butyric acid (IBA) have been isolated from plants. NAA (naphthalene acetic acid) and 2, 4-D (2, 4-dichlorophenoxyacetic) are synthetic auxins. They help to initiate rooting in stem cuttings, an application widely used for plant propagation. Auxins promote flowering e.g. in pineapples. They help to prevent fruit and leaf drop at early stages.

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131. (3)

The thymus is a specialized lobular organ of the immune system located in the upper chest, specifically between the lungs and behind the sternum. It plays a vital role in the maturation of T-lymphocytes. Interestingly, the thymus is quite large at birth but gradually shrinks in size after puberty, which is why immune responses in elderly individuals tend to be weaker.

(NCERT Ed. 2025-26, 11th, Page No. 243)

132. (4)

The human respiratory system functions through coordinated phases. The conducting part (nose to bronchioles) filters and humidifies air, while the exchange part (alveoli) is where gas diffusion occurs. A healthy adult breathes about 12-16 times per minute. However, expiration is a passive process where air is pushed out because intra-pulmonary pressure becomes higher than atmospheric pressure, and it is CO_2, not O_2, that has a significantly higher solubility in blood.

(NCERT Ed. 2025-26, 11th, Page No. 185, 186)

133. (2)

When a living tissue is ground in trichloroacetic acid and filtered, the acid-soluble pool (filtrate) contains thousands of small organic compounds with molecular weights ranging from 18 to 800 Daltons. This pool primarily represents the cytoplasmic composition, including amino acids, sugars, and nucleotides. In contrast, the acid-insoluble fraction contains large macromolecules like proteins and nucleic acids, as well as membrane fragments.

(NCERT Ed. 2025-26, 11th, Page No. 109)

134. (4)

The organism that causes white spots on mustard leaves is *Albugo.* It belongs to class Phycomycetes and has an aseptate and coenocytic mycelium. It is a multicellular, filamentous (mycelial) fungus. Basidiospores are characteristic of the class Basidiomycetes. In Phycomycetes, asexual reproduction occurs via zoospores or aplanospores produced endogenously in sporangia. Conidia are the asexual spores characteristic of Ascomycetes and Deuteromycetes.

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135. (1)

Pea is a member of family Leguminosae.

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[Contd...

136. (3)

The diagram provided shows fibers (collagen) and cells (fibroblasts) oriented in an irregular and random pattern. This is a characteristic of dense irregular connective tissue. This specific type of tissue is designed to provide strength and resist tearing in multiple directions, and its most common location in the human body is the dermis of the skin.

(NCERT Ed. 2020-21, 11th, Page No. 103)

137. (3)

In the dihybrid cross, the phenotypes round, yellow; wrinkled, yellow; round, green and wrinkled, green appeared in the ratio 9:3:3:1. Based upon such observations on dihybrid crosses (crosses between plants differing in two traits) Mendel proposed a second set of generalisations that we call Mendel's Law of Independent Assortment. The law states that 'when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters'.

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138. (4)

Organism		Order
Petunia	-	Polymoniales
Triticum	-	Poales
Panthera	-	Carnivora
Gorilla	-	Primata

(NCERT Ed. 2022-23, 11th, Page No. 7)

139. (1)

The biogas plant consists of a concrete tank (10-15 feet deep) in which bio-wastes are collected and a slurry of dung is fed. A floating cover is placed over the slurry, which keeps on rising as the gas is produced in the tank due to the microbial activity. The biogas plant has an outlet, which is connected to a pipe to supply biogas to nearby houses. The spent slurry is removed through another outlet and may be used as fertiliser.

(NCERT Ed. 2025-26, 12th, Page No. 156)

140. (1)

Muscle fibers have specific structural and functional characteristics. A sarcomere is the functional unit of contraction defined as the segment between two Z lines. The H-zone represents the central portion of the thick (myosin) filament that is not overlapped by thin (actin) filaments. While red muscle fibers have many mitochondria to support aerobic activity, white muscle fibers have fewer mitochondria but possess a very extensive sarcoplasmic reticulum for rapid calcium release during fast-twitch contractions.

(NCERT Ed. 2025-26, 11th, Page No. 220, 223)

141. (2)

Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.

(NCERT Ed. 2025-26, 12th, Page No. 152)

142. (3)

The anatomy of a cockroach is highly segmented. While the body is divided into the head, thorax, and abdomen, the head itself is a complex structure formed by the fusion of six embryonic segments. The exoskeleton consists of hardened plates called sclerites. Both sexes have anal cerci for sensing, but only males possess the short, thread-like anal styles which help in distinguishing them from females.

(NCERT Ed. 2020-21, 11th, Page No. 112)

143. (2)

In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy.

(NCERT Ed. 2025-26, 12th, Page No. 15)

144. (3)

The green plants (producers) in a terrestrial ecosystem capture only about 1% of the energy of sunlight that falls on their leaves and convert it into food energy.

Incident Sunlight: 100,000 J

Energy captured by Producers: 1% of 100,000 J = 1,000 J

According to the 10% law of energy transfer, only 10% of the energy available at one trophic level is transferred to the next.

Energy at Producer level (T1): 1,000 J

Energy reaching Primary Consumers (T2): 10% of 1,000 J = 100 J

Primary Consumer to Secondary Consumer (10% Law)

Energy at Primary Consumer level (T2): 100 J

Energy reaching Secondary Consumers (T3): 10% of 100 J = 10 J

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[Contd...

145. (4)

The C_4 plants respond to higher temperatures and show higher rate of photosynthesis while C_3 plants have a much lower temperature optimum. Tropical plants have a higher temperature optimum than the plants adapted to temperate climates.

(NCERT Ed. 2025-26, 11th, Page No. 150)

146. (3)

The human heart is a complex muscular organ derived from the mesoderm. It is protected by a double-walled pericardium and controlled by the SA node (pacemaker). However, it is the left atrium that receives oxygenated blood from the lungs via the pulmonary veins. Deoxygenated blood from the body actually enters the right atrium through the vena cava, making the statement about the left atrium receiving deoxygenated blood incorrect.

(NCERT Ed. 2025-26, 11th, Page No. 198)

147. (3)

In a typical neuron, the cell body and dendrites contain granular bodies called Nissl's granules, which are actually sites of protein synthesis (rough endoplasmic reticulum). However, these granules are completely absent in the axon. The axon is specialized for the conduction of nerve impulses away from the cell body and does not participate in the protein synthesis typical of the rest of the neuron.

(NCERT Ed. 2025-26, 11th, Page No. 231)

148. (1)

The human kidney is incredibly efficient at water conservation. Through the counter-current mechanism—which involves the specific flow of filtrate in Henle's loop and blood in the vasa recta—a high concentration of salts is maintained in the inner medulla of the kidney. this "osmotic gradient" allows the kidney to reabsorb massive amounts of water, ultimately producing urine that is up to four times more concentrated than the initial filtrate.

(NCERT Ed. 2025-26, 11th, Page No. 211, 212)

149. (2)

Genetic engineering or recombinant DNA technology can be accomplished only if we have the key tools, i.e., restriction enzymes, polymerase enzymes, ligases, vectors and the host organism. The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase, which acts on cut DNA molecules and joins their ends. When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky-ends' and, these can be joined together (end-to-end) using DNA ligases. During DNA replication, on one strand (the template with polarity 3′→5′), the replication is continuous, while on the other (the template with polarity 5′→3′), it is discontinuous. The discontinuously synthesised fragments are later joined by the enzyme DNA ligase

(NCERT Ed. 2025-26, 12th, Page No. 90)

150. (3)

The LH surge around Day 13–14 is the key trigger for ovulation. If this surge is artificially inhibited:

- Ovulation will not occur
- The ruptured Graafian follicle will not form a functional corpus luteum
- Therefore, progesterone will not rise during Days 15–25
- Without progesterone, the endometrium does not properly convert into the secretory phase

So the most likely event in the luteal-phase window is failure of progesterone rise and failure of secretory transformation of the uterine lining.

(NCERT Ed. 2025-26, 12th, Page No. 34)

151. (3)

There is a clear division of labour within the chloroplast. The membrane system is responsible for trapping the light energy and also for the synthesis of ATP and NADPH. In stroma, enzymatic reactions synthesise sugar, which in turn forms starch. The former set of reactions, since they are directly light driven are called light reactions (photochemical reactions). The latter are not directly light driven but are dependent on the products of light reactions (ATP and NADPH). Hence, to distinguish the latter they are called, by convention, as dark reactions (carbon reactions). However, this should not be construed to mean that they occur in darkness or that they are not light-dependent.

(NCERT Ed. 2025-26, 11th, Page No. 136)

152. (3)

Enzymes are biological catalysts with very specific requirements. They work best at an "optimum" temperature and pH, and their activity drops if these conditions change. Competitive inhibitors are molecules that "mimic" the substrate and compete for the active site. While increasing substrate concentration initially speeds up a reaction, it eventually hits a maximum velocity (Vmax) where all enzyme sites are saturated, and adding more substrate will not increase the rate further.

(NCERT Ed. 2025-26, 11th, Page No. 116, 117)

[Contd...

153. (2)

Bacteria that live in extreme salty areas are called halophiles.

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154. (2)

To find the intrinsic rate of natural increase (r) for this population, we use the *Verhulst-Pearl Logistic Growth* formula, as the mention of an "asymptote" indicates that the population is following a logistic growth curve where it is approaching its carrying capacity.

$$\frac{dN}{dt} = rN \times \frac{K - N}{K}$$

$$0.5 = r\,(200) \times \frac{260 - 200}{260}$$

Therefore, $r = \frac{6.5}{60} = 0.0108$

(NCERT Ed. 2025-26, 12th, Page No. 195)

155. (2)

When a phosphate group is linked to OH of 5' C of a nucleoside through phosphoester linkage, a corresponding nucleotide (or deoxynucleotide depending upon the type of sugar present) is formed. Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide. More nucleotides can be joined in such a manner to form a polynucleotide chain. A polymer thus formed has at one end a free phosphate moiety at 5' -end of sugar, which is referred to as 5'-end of polynucleotide chain. Similarly, at the other end of the polymer the sugar has a free OH of 3'C group which is referred to as 3' -end of the polynucleotide chain. The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. The two chains are coiled in a right-handed fashion.

(NCERT Ed. 2025-26, 12th, Page No. 81)

156. (1)

List-I	List-II
Hydrolases	Cleavage of bonds by adding water
Ligases	Joining of two molecules
Transferases	Transfer of a group (other than H)
Oxidoreductases	Catalysis of oxidation-reduction

(NCERT Ed. 2025-26, 11th, Page No. 117)

157. (2)

List-I (Animal)	List-II (Distinguishing feature)
Chameleon	Oviparous with direct development
Salamandra	Oviparous with indirect development
Neophron	Air sacs connected to lungs
Macropus	Presence of milk producing glands

(NCERT Ed. 2025-26, 11th, Page No. 48, 49, 50)

158. (3)

Immunity can be either active (the body makes its own antibodies) or passive (ready-made antibodies are given). When a person is bitten by a snake, there is no time for their own immune system to respond. Therefore, doctors inject pre-formed antibodies (antitoxin) directly into the patient. Because the person's own body did not produce these antibodies, this is a classic example of passive immunity, providing immediate but short-term protection.

(NCERT Ed. 2025-26, 12th, Page No. 136)

159. (4)

The brainstem is the posterior part of the brain that connects the cerebrum with the spinal cord. It is composed of three main regions: the midbrain, the pons, and the medulla oblongata. The cerebrum, which is the largest part of the brain responsible for higher-order functions, is situated above the brainstem and is not considered a part of it.

(NCERT Ed. 2025-26, 11th, Page No. 236)

160. (1)

The common Indian frog has several distinct anatomical features. Its heart is three-chambered, consisting of two auricles (atria) and one single ventricle. The other statements are incorrect because the frog's kidneys are located posteriorly (in the back), its skin is always moist and slippery to aid in breathing, and its eyes are protected by a transparent "nictitating membrane" while in water, not the tympanum (which is the external ear).

(NCERT Ed. 2025-26, 11th, Page No. 82)

161. (3)

When $FADH_2$ donates its electrons to the Electron Transport System (ETS), it enters at Complex II (Succinate Dehydrogenase). From this point, the complexes that facilitate proton movement (pumping) from the mitochondrial matrix to the intermembrane space are Complex III and Complex IV.

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[Contd...

162. (3)

Lymphoid organs are the sites where immune cells are produced or interact with pathogens. Primary organs (bone marrow and thymus) are where lymphocytes mature. Secondary organs (spleen, lymph nodes, MALT) are where they actually encounter and fight antigens. MALT (Mucosa-Associated Lymphoid Tissue), located in the linings of the digestive, respiratory, and urogenital tracts, is very extensive and accounts for approximately 50% of all lymphoid tissue in the body.

(NCERT Ed. 2025-26, 12th, Page No. 138)

163. (4)

Human blood groups are determined by antigens on the surface of RBCs. Individuals with blood group 'O' have no A or B antigens on their RBCs, meaning their blood will not be attacked by the antibodies of a recipient with a different group. This makes 'O' group individuals universal donors who can safely give blood to anyone. Conversely, group AB individuals are universal recipients because they lack the antibodies that would attack donated A or B blood.

(NCERT Ed. 2025-26, 11th, Page No. 195)

164. (3)

In prokaryotes, such as, *E. coli*, though they do not have a defined nucleus, the DNA is not scattered throughout the cell. DNA (being negatively charged) is held with some proteins (that have positive charges) in a region termed as 'nucleoid'. In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.

(NCERT Ed. 2025-26, 12th, Page No. 83)

165. (3)

Pyramid of energy is always upright, can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.

(NCERT Ed. 2025-26, 12th, Page No. 213)

166. (2)

Contraceptive methods work by preventing the meeting of sperm and egg. Diaphragms and cervical caps are reusable rubber barriers that are inserted into the female reproductive tract to cover the cervix; they are much more effective when used with spermicidal creams. Unlike condoms, which are often used once, these barriers can be cleaned and reused. Nirodh is specifically a popular brand of male condom, not an intrauterine device (IUD).

(NCERT Ed. 2025-26, 12th, Page No. 44)

167. (3)

Plant tissue culture relies on the ability of plant cells to regenerate into whole plants (totipotency). Meristem culture is a specific technique used to obtain virus-free plants, because the apical and axillary meristems of a plant are usually free of viruses even if the rest of the plant is infected. Therefore, the statement that "all plant tissues always harbour virus" is incorrect, as the rapid cell division in meristems often outpaces the spread of the virus.

(NCERT Ed. 2025-26, 12th, Page No. 178)

168. (1)

In a few plant species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits. There are several records of very old yet viable seeds. The oldest is that of a lupine, *Lupinus arcticus* excavated from Arctic Tundra. The seed germinated and flowered after an estimated record of 10,000 years of dormancy. A recent record of 2000 years old viable seed is of the date palm, *Phoenix dactylifera* discovered during the archeological excavation at King Herod's palace near the Dead Sea. The number of ovules in an ovary may be one (wheat, paddy, mango) to many (papaya, water melon, orchids).

(NCERT Ed. 2025-26, 12th, Page No. 7)

169. (1)

During the process of muscle contraction, the distance between the Z lines decreases as the sarcomere shortens. The I-bands (thin filaments only) get reduced in length because they are pulled toward the center. However, the A-bands (thick filaments) retain their original length because they do not shrink; instead, the thin filaments simply slide over the thick filaments, which is the fundamental concept of the sliding filament theory.

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170. (2)

List-I (Treatment method)	List-II (Mechanism Description)
Radiotherapy	Tumor cells are lethally irradiated while protecting normal tissue
Chemotherapy	Use of drugs to kill cancer cells; often has side effects like hair loss
Immunotherapy	Use of biological response modifiers like alpha-interferon
Surgery	Surgical removal of localized cancerous mass

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[Contd...

171. (2)

DNA replication S phase: Only the amount of DNA doubles, not the number of chromosomes.

Disintegration of nuclear envelope: This is a structural change to allow the spindle fibers to reach the chromosomes; it has no effect on chromosome count.

Recombination in pachytene: This involves the exchange of genetic material between non-sister chromatids of homologous chromosomes. It changes the genetic combination but not the number of chromosomes.

Anaphase: This is the stage where the centromeres split and the sister chromatids separate. As soon as the sister chromatids separate, each one is considered an independent daughter chromosome. Because the cell has not yet divided into two (cytokinesis hasn't happened), all these separated chromosomes are still within the single parent cell. Therefore, the chromosome number is temporarily doubled until the cell completes division.

(NCERT Ed. 2025-26, 11th, Page No. 123)

172. (3)

List-I	List-II
Squamous epithelium	Present in walls of blood vessels
Tendon	Connect muscle to bones
Ligament	Connect bone to bone
Cuboidal epithelium	Present in ducts of glands

(NCERT Ed. 2020-21, 11th, Page No. 101, 103)

173. (1)

Calvin proposed that plants change light energy to chemical energy by transferring an electron in an organised array of pigment molecules and other substances.

(NCERT Ed. 2025-26, 11th, Page No. 132)

174. (3)

Diatoms are the chief 'producers' in the oceans. In diatoms the cell walls form two thin overlapping shells, which fit together as in a soap box. The walls are embedded with silica and thus the walls are indestructible. Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as ‘diatomaceous earth’. All protozoans are heterotrophs and live as predators or parasites. They are believed to be primitive relatives of animals. Sporozoans include diverse organisms that have an infectious spore-like stage in their life cycle.

(NCERT Ed. 2025-26, 11th, Page No. 14)

175. (1)

The smooth endoplasmic reticulum is the major site for synthesis of lipid. In animal cells lipid-like steroidal hormones are synthesised in SER.

(NCERT Ed. 2025-26, 11th, Page No. 95)

176. (1)

Prophase of the first meiotic division has been further subdivided into the following five phases based on chromosomal behaviour, i.e., Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis.

(NCERT Ed. 2025-26, 11th, Page No. 126)

177. (2)

During the light reaction of photosynthesis, the proton accumulation is towards the inside of the thylakoid membrane, i.e., in the lumen.

(NCERT Ed. 2025-26, 11th, Page No. 140)

178. (4)

B. Meiosis in sporangial cells: In the diploid sporophyte (the dominant plant body), specialized cells in the sporangia undergo meiosis to reduce the chromosome number.

D. Production of spores: As a result of meiosis, haploid spores are produced and released.

A. Formation of thalloid gametophyte: The spores germinate in damp, shady places to form a multicellular, free-living, photosynthetic haploid gametophyte called a prothallus.

C. Production of gametes: The gametophyte bears sex organs (antheridia and archegonia) that produce male and female gametes through mitosis.

E. Formation of zygote: The male gamete (antherozoid) fuses with the female gamete (egg) in the presence of water (fertilization) to form a diploid zygote, which eventually grows back into a sporophyte.

(NCERT Ed. 2025-26, 11th, Page No. 30)

179. (1)

The substrate-level phosphorylation (SLP) steps are those in which ATP or GTP is directly synthesized from a high-energy substrate without the use of an electron transport chain.

From the given options, the substrate-level phosphorylation steps are:

Phosphoenolpyruvate → Pyruvate

Succinyl-CoA → Succinic acid

(NCERT Ed. 2025-26, 11th, Page No. 159)

180. (1)

Transgenic animals are created by inserting foreign genes into their DNA, which serves many purposes in research. They are used to study how genes regulate normal growth, to produce valuable human proteins (like alpha-1-antitrypsin for emphysema), and to test the safety of new vaccines. However, they are not created to "ensure they never develop a disease"; in fact, researchers often create transgenic models that are *more* susceptible to diseases like cancer or Alzheimer's so they can study the progression and potential cures.

(NCERT Ed. 2025-26, 12th, Page No. 184)

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